A Reduced-Scale Reproduction of Language-Model Memorization Experiments

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***Abstract*—Language models can store information about individual training examples, but natural text mixes memorization with generalization. This paper presents a reduced-scale PyTorch reproduction of the synthetic-data memorization experiments of Morris et al. [1]. A causal Transformer with 104,514 trainable parameters is trained for 100 epochs on uniformly random 64-token binary sequences (1,000 to 10,000 sequences). Memorization is estimated as the reduction of training-sequence negative log-likelihood (NLL) below the 64-bit uniform reference. Estimated memorization rises from 797.04 to 4,868.55 bits (0.007626 to 0.046583 bits/parameter) as the dataset grows, while evaluation NLL stays above 1 bit/token and final training NLL falls below it, indicating overfitting. Exploratory 30-epoch ablations show that total memorization increases with model size (365.08 to 552.36 bits) while bits/parameter decreases, that dropout $p = 0.1$ lowers the estimate by 7.3% in a single run, and that three random seeds give 488.39 $\pm$ 18.41 bits (mean $\pm$ sample standard deviation). Training loss was still decreasing at epoch 100, no capacity plateau was observed, and the approximately 3.6 bits/parameter estimate of [1] was not reproduced. The study is a limited reproduction of the measurement approach, not of the original scaling law.**

***Index Terms*—language-model memorization, Transformer, random binary sequences, negative log-likelihood, bits per parameter, PyTorch.**

I. INTRODUCTION

Language models can encode information about their training examples, but predictive performance on natural text mixes information about individual examples with information about the data-generating distribution. Morris et al. [1, 2] separate unintended memorization from generalization by training GPT-style models on synthetic, uniformly random sequences, for which no useful generalization is possible. In that setting, any reduction of a sequence's code length below its entropy must come from information stored about that particular sequence. Their ICML 2026 study reports a capacity of approximately 3.6 bits per parameter and observes that memorization saturates once the dataset carries more information than the model can store [1].

This project implements a compact decoder-only Transformer [3] in PyTorch under a single Google Colab Tesla T4 budget. Random binary sequences of length 64 are used with dataset-size, model-size, dropout, and random-seed experiments. The model and training budget are far smaller than in the original work, so the aim is to study the behaviour of the measurement approach and its trends, not to claim an exact reproduction. All numbers reported here are taken from, or recomputed from, the saved output files of a single experiment run.

II. METHODOLOGY

A. Synthetic Data and Reference Entropy

Each example is an independently sampled sequence $x = (x_1, \ldots, x_T)$ with $T = 64$ i.i.d. uniform binary tokens. Its entropy is $T \log_2 2 = 64$ bits, so a model that outputs a uniform distribution has a sequence NLL of 64 bits, or 1 bit/token. Because the sequences contain no reusable structure, a training-set NLL below this reference is treated as a proxy for information stored about the training examples. A single 10,000-sequence dataset is generated from seed 2026, and the training sets of 1,000, 2,000, and 5,000 sequences are its prefixes, so the dataset-size comparison is nested. A separate evaluation set of 2,000 sequences is generated from seed 9090 and was checked to differ from the first 2,000 training sequences. The model reads $[\mathrm{BOS}, x_1, \ldots, x_{63}]$ and predicts $[x_1, \ldots, x_{64}]$, so each sequence has 64 prediction targets.

B. Transformer Architecture

The model is a GPT-style causal Transformer with learned token embeddings (two binary symbols plus an input-only beginning-of-sequence (BOS) token), learned positional embeddings, pre-layer-normalization blocks with masked multi-head self-attention and a GELU feed-forward network, a final layer normalization, and a two-class next-token output head. The main (medium) configuration has embedding dimension 64, 2 layers, 4 attention heads, feed-forward dimension 256 (four times the embedding dimension), and 104,514 trainable parameters. The small and large variants use embedding dimensions of 32 and 128 with the same number of layers and heads, giving 27,682 and 405,634 trainable parameters, respectively. Dropout, when used, is applied to attention weights, attention output, and feed-forward output.

C. Training Configuration

Models are trained with cross-entropy loss, the AdamW optimizer [4] with learning rate 3×10^{-4} (PyTorch default settings, no learning-rate schedule), and batch size 128 with shuffled mini-batches, using PyTorch 2.11.0 on a Tesla T4 GPU in Google Colab. The seed controls model initialization and mini-batch order. The *main dataset-size sweep* trains the medium model for 100 epochs on each dataset size (seed 2026) and reports the *final-epoch* model. The *ablations* (model size, dropout, and random seed) all use 5,000 training sequences and 30 epochs per run, with seeds 123 (model size, dropout), and 123, 456, and 789 (seed sensitivity). For the ablations the reported checkpoint is the epoch with the *lowest evaluation NLL*. The two procedures there-

fore differ in both training length and checkpoint-selection policy; this matters for comparing tables (Section IV).

D. Memorization Metric

For a sequence x_i with model distribution p_θ, the sequence NLL in bits is

$$\mathrm{NLL}_{\mathrm{bits}}(x_i) = -\sum_{t=1}^{T} \log_2 p_\theta(x_{i,t} \mid x_{i,<t}). \tag{1}$$

Relative to the uniform reference of $T = 64$ bits, the per-sequence contribution is

$$m_i = \max\bigl(0,\ 64 - \mathrm{NLL}_{\mathrm{bits}}(x_i)\bigr), \tag{2}$$

and the total estimated memorization of a training set D is

$$M(D) = \sum_{x_i \in D} m_i \quad \text{(bits)}. \tag{3}$$

The normalized statistic is $M(D)/P$ in bits/parameter, where P is the number of trainable parameters. The positive part is applied per sequence before summation, and $M(D)$ is computed on the training sequences. This is an empirical, implementation-specific estimate under a uniform reference. It is not a universal definition of memorization and not a proven capacity limit. Because of the per-sequence clipping, m_i is non-negative even when the model has not learned anything about x_i, so $M(D)$ contains a positive noise component (quantified in Section III-A).

E. Implementation and Sanity Checks

The notebook includes the following checks, all of which passed. A uniform-output model on 32 random sequences yields exactly 64 bits per sequence and zero estimated memorization. In a causal-leakage test, changing the last four tokens of an 8-token input changed the logits at the first four positions by a maximum of 0 (tolerance 10^{-6}). The medium model has exactly 104,514 trainable parameters, and the parameter counts of all three variants (27,682, 104,514, and 405,634) also follow from the architecture. Dataset shapes and binary-token constraints are asserted, and the final integrity cell checks run counts, finite values, completion of 100 epochs in the main sweep, and the bits/parameter arithmetic. Recomputing $M(D)$ and bits/parameter from the result CSVs agrees with the stored values, and the three runs that share one configuration (medium model, no dropout, seed 123, 5,000 sequences) agree on estimated memorization to within 0.001 bits.

III. Experimental Results

A. Dataset-Size Sweep

Table I and Fig. 1 report the 100-epoch sweep. Estimated memorization increases from 797.04 bits at 1,000 sequences to 4,868.55 bits at 10,000 sequences, and the normalized estimate from 0.007626 to 0.046583 bits/parameter. No plateau appears in the tested range. Final training NLL (0.991275 to 0.997056 bits/token) is below the 1-bit/token reference for all four runs, whereas evaluation NLL (1.003021 to 1.007238 bits/token) is above it, consistent with overfitting.

The growth is not proportional to dataset size. Average memorization per training sequence is 0.80, 0.78, 0.41, and 0.49 bits for 1,000, 2,000, 5,000, and 10,000 sequences (calculated from Table I), i.e. below 1 bit out of 64, and it is non-monotonic. With one seed per size, it cannot be determined whether this reflects the training state at epoch 100 or run-to-run variability. The final column of Table I applies the same estimator to the 2,000 held-out sequences, which the model has not been trained on. It yields 449.85 to 559.96 bits (about 0.22 to 0.28 bits per sequence), confirming that the clipped estimator has a non-zero noise floor and that part of each training-set value is not memorization of those sequences.

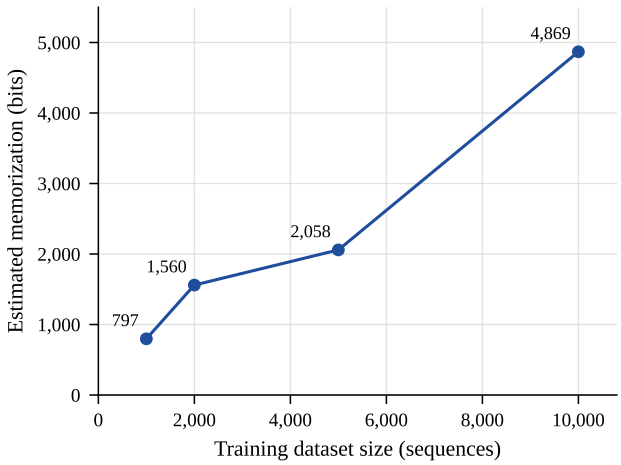


Fig. 1. Estimated memorization versus training dataset size for the medium model (100 epochs, final-epoch model). No capacity plateau appears in the tested range.

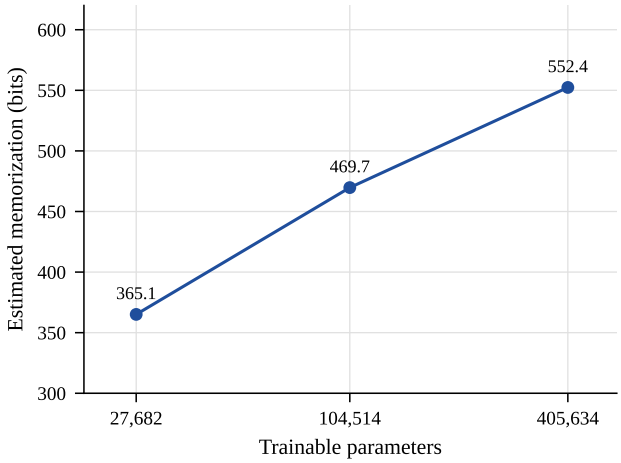


Fig. 2. Total estimated memorization versus trainable parameters (5,000 sequences, 30 epochs, lowest-evaluation-NLL checkpoint; log-scaled x axis).

B. Model-Size Ablation

On 5,000 sequences trained for 30 epochs (Table II, Fig. 2), total estimated memorization increases from 365.08 bits for the 27,682-parameter model to 552.36 bits for the 405,634-parameter model, while bits/parameter decreases from 0.013188 to 0.001362. The comparison is exploratory: each variant has one seed, the models may not be in comparable training states after 30 epochs, and the checkpoint epochs differ (17, 20, and 20).

C. Dropout Ablation

Adding dropout $p = 0.1$ changes the estimated memorization from 469.73 to 435.34 bits, a reduction of 34.39 bits (7.3%) in this single-seed comparison (Table III, Fig. 3).

TABLE I
DATASET-SIZE SWEEP (MEDIUM MODEL, 104,514 PARAMETERS, 100 EPOCHS, FINAL-EPOCH MODEL, SEED 2026)

Dataset size (sequences)	Training NLL (bits/token)	Evaluation NLL (bits/token)	Est. memorization (bits)	Memorization (bits/parameter)	Held-out diagnostic† (bits)	Epochs completed
1,000	0.991275	1.007238	797.04	0.007626	539.54	100
2,000	0.991564	1.006584	1,559.70	0.014923	559.96	100
5,000	0.997056	1.003021	2,058.11	0.019692	449.85	100
10,000	0.995656	1.003504	4,868.55	0.046583	470.05	100

†Same estimator applied to the 2,000 evaluation sequences; a noise-floor diagnostic over a different number of sequences, not directly comparable in magnitude with the training-set estimate.

TABLE II
MODEL-SIZE ABLATION (5,000 SEQUENCES, 30 EPOCHS, SEED 123, LOWEST-EVALUATION-NLL CHECKPOINT)

Variant	Parameters	Eval. NLL (bits/token)	Mem. (bits)	Mem. (bits/param.)	Best epoch
Small	27,682	1.000186	365.08	0.013188	17
Medium	104,514	1.000286	469.73	0.004494	20
Large	405,634	1.000344	552.36	0.001362	20

TABLE III
DROPOUT ABLATION (MEDIUM MODEL, 5,000 SEQUENCES, 30 EPOCHS, SEED 123)

Dropout (probability)	Eval. NLL (bits/token)	Mem. (bits)	Mem. (bits/param.)	Best epoch
0.0	1.000286	469.73	0.004494	20
0.1	1.000244	435.34	0.004165	20

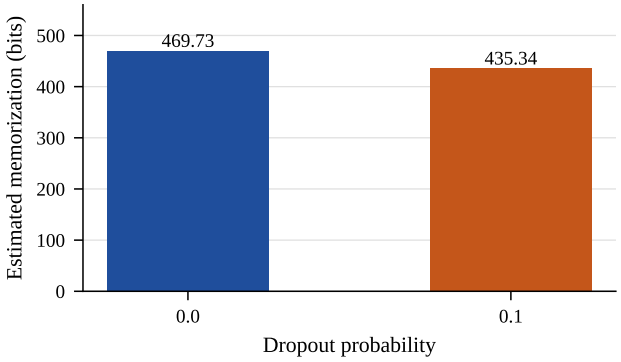


Fig. 3. Estimated memorization for dropout probabilities 0.0 and 0.1 (single seed per setting).

Evaluation NLL is nearly unchanged (1.000286 versus 1.000244 bits/token). The difference is of the same order as the spread among random seeds in Section III-D (469.73 to 506.54 bits), so it should not be interpreted as a robust causal effect of dropout.

D. Random-Seed Sensitivity

Across seeds 123, 456, and 789 (Table IV), estimated memorization has mean 488.39 bits and sample standard deviation 18.41 bits ($n = 3$); normalized, the mean is 0.004673 bits/parameter with standard deviation 0.000176. The seed changes initialization and mini-batch order while the 5,000 training sequences and the evaluation set are fixed, so this measures training-seed variability only, not variation across independently sampled datasets. Three runs are too few for confidence intervals, so this is only an indication of run-to-run variability at 5,000 sequences and 30 epochs.

E. Training Dynamics

Fig. 4 shows training and evaluation NLL over the 100 epochs of the main sweep. Evaluation NLL reaches its minimum early (epoch 14, 9, 9, and 15 for 1,000 to 10,000 sequences) with values of at least 1.000241 bits/token, i.e. never below the uniform baseline, and then drifts upward to the final values in Table I. The drift is largest for the 1,000- and 2,000-sequence runs. Training NLL continues to decrease through the final epoch in all four runs (the mean of the last 10 epochs is lower than that of the preceding 10 epochs), so full convergence was not established. The training curve in Fig. 4 is the running average of the mini-batch loss during each epoch, so it differs slightly from the post-training evaluation in Table I (for example, 0.991854 versus 0.991275 bits/token training NLL at 1,000 sequences).

TABLE IV
RANDOM-SEED SENSITIVITY (MEDIUM MODEL, 5,000 SEQUENCES, 30 EPOCHS)

Seed	Eval. NLL (bits/token)	Mem. (bits)	Mem. (bits/param.)	Best epoch
123	1.000286	469.73	0.004494	20
456	1.000288	506.54	0.004847	10
789	1.000306	488.92	0.004678	7
Mean	1.000293	488.39	0.004673	—
Std. dev.	0.000011	18.41	0.000176	—

IV. DISCUSSION AND LIMITATIONS

What the experiments support. The implementation behaves as expected for random data: training NLL can be pushed below the 1-bit/token reference while held-out NLL never goes below it, and the estimated memorization grows with dataset size over the tested range (Table I). The measured 0.046583 bits/parameter is about 1.3% of the approximately 3.6 bits/parameter reported in [1]. The model has stored only 4,868.55 of the 640,000 bits of entropy in the 10,000-sequence training set, so it is far from any saturation regime; the absence of a plateau here says nothing about whether one would appear at larger scale.

What they do not support. The results do not reproduce the original capacity estimate or plateau, and do not establish a scaling law in dataset size or parameters. The model-size, dropout, and seed results rest on one to three runs of 30 epochs each. In addition, the clipped estimator has a positive noise floor (Section III-A), so total estimated memorization overstates the information stored about the training sequences.

Differences between procedures. The main sweep reports the final-epoch model after 100 epochs, while the ablations use 30 epochs and select the epoch with the lowest evaluation NLL. That selection uses the same 2,000 evaluation sequences used for reporting, so the ablation evaluation NLL values are optimistic. The two procedures give very different memorization for the same nominal configuration

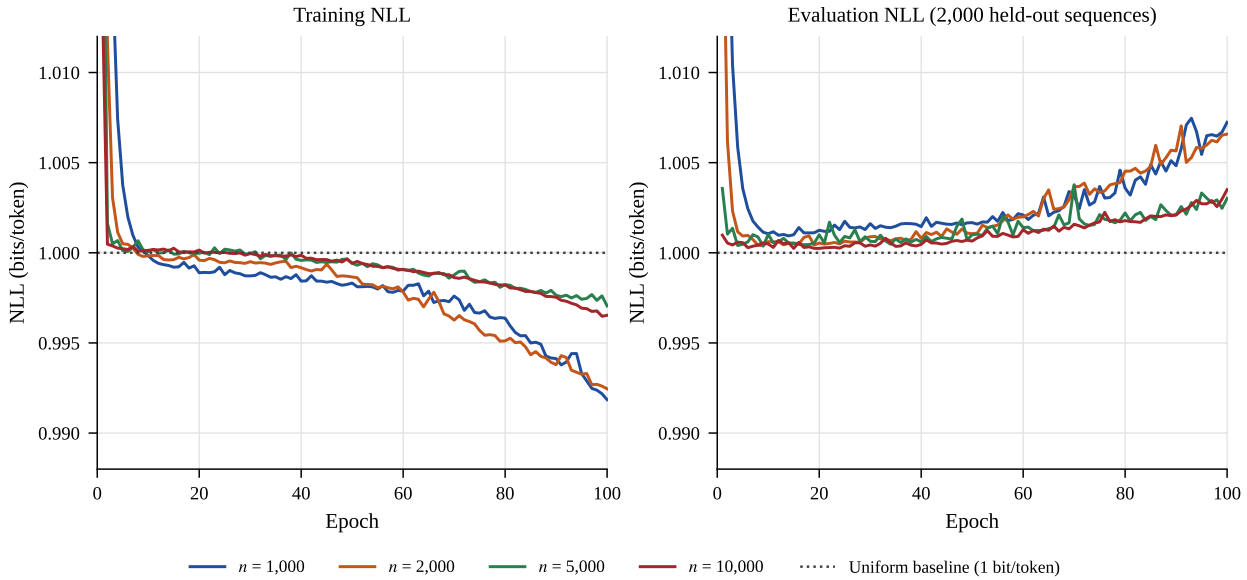


Fig. 4. Training and evaluation dynamics of the main dataset-size sweep (100 epochs, per-epoch logged values). Training NLL (left) falls below the 1 bit/token uniform baseline while evaluation NLL (right) drifts above it; the y axis is truncated to the range 0.988–1.012 bits/token, which clips the first epoch of some runs.

(medium model, 5,000 sequences): 2,058.11 bits in Table I versus 469.73 bits in Tables II–IV. Values from the ablation tables must therefore not be compared with Table I.

Remaining limitations. (1) The model is much smaller than those of the original study, and training is short: training loss was still decreasing and evaluation loss drifting upward at epoch 100. (2) Convergence was not demonstrated. (3) The main sweep has one seed per dataset size. (4) Variation across independently sampled datasets was not measured. (5) The estimator uses a uniform reference and per-sequence clipping. Future work should use larger models, longer training with explicit convergence criteria, a separate validation set for checkpoint selection, multiple seeds and datasets, and a test of whether total memorization plateaus when dataset entropy exceeds model capacity.

V. CONCLUSION

A compact causal Transformer was trained on uniformly random binary sequences, and memorization was estimated from the reduction in training-sequence NLL below the 64-bit uniform reference. Estimated memorization increased from 797.04 to 4,868.55 bits as the dataset grew from 1,000 to 10,000 sequences, training NLL fell below 1 bit/token, and evaluation NLL stayed above it. The 30-epoch ablations suggest that total memorization grows with model size but bits/parameter falls, and that dropout $p = 0.1$ and random seed changes move the estimate by amounts that are comparable to each other. No capacity plateau was observed, training had not converged, and the approximately 3.6 bits/parameter estimate was not reproduced. This work is best described as a reduced-scale implementation and partial reproduction of the measurement approach.

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